Vegetable Resources

Appendix of Useful Tables

Table 1. Annual average water use and critical growth stage of vegetable crops in Texas.

Crop	Inches/A	Critical Need Stage
Asparagus	10-18	establishment and fern development
Bean, green	10-15	bloom and pod set
Bean, pinto	15-20	bloom and pod set
Beet, table	10-15	establishment and early growth
Broccoli	20-25	establishment and heading
Cabbage	20-30	uniform throughout growth
Cantaloupe	13-20	establishment vining to first net
Carrot	10-15	emergence through establishment
Cauliflower	20-30	establishment and 6-7 leaf stage
Celery	30-35	uniform, last month of growth
Collards/Kale	12-14	uniform throughout growth
Corn, sweet	20-35	establishment, tassel elongation, ear development
Cowpea	10-15	bloom, fruit set, pod development
Cucumber, pickle	15-20	establishment, vining, fruit set
Cucumber, slicer	20-25	establishment, vining, fruit set
Eggplant	20-35	bloom through fruit set
Garlic	15-20	rapid growth to maturity
Lettuce	8-12	establishment
Mustard green	10-15	uniform throughout growth
Okra	15-20	uniform throughout growth
Onion	25-30	establishment, bulbing to maturity
Pepper, bell	25-35	establishment, bloom set
Pepper, jalapeno	25-30	uniform throughout growth
Potato	20-40	vining, bloom, tuber initiation
Pumpkin	25-30	2-4 wk. after emergence, bloom, fruit set and development
Radish, red globe	5-10	rapid growth and development
Spinach	10-15	uniform throughout growth, after each cut if needed
Squash	7-10	uniform throughout growth
Sweetpotato	10-20	uniform until 2-3 wk. prior to anticipated harvest
Tomato	20-25	bloom through harvest
Turnip	10-15	uniform throughout growth

Watermelon	10-15	uniform until 10-14 days prior to anticipated harvest

Table 2. Suggested varieties for Texas.

Crop	Suggestions
Asparagus	UC 157, UC 72, UC 500W, UC 72, Jersey Gem, Jersey Giant, Jersey Centennial
Bean	Green: Benchmark, Blue Lake 274, Derby, Jade, Landmark, Opus, Strike, Flat pod: Calgreen, Magnum, Roma II Pinto: Cinnabar, Bill Z., Fiesta, Othello, Pinata III, Pinray, U.I. 126
Beet, table	Detroit Dark Red S.T., Red Ace, Red Cloud, Warrior
Broccoli	Everst, Heritage, Liberty, Sultan, Marathon, Patriot, Signal, Triathlon
Cabbage	Blue Vantage, Bravo, Cheers, Emblem, Fortress, Pennant, Solid Blue 790, Solid Blue 760, Vantage Point, Blue Thunder Red Type: Cardinal, Red Jewel, Red Rock, Red Rookie, Rio Grande Red
Cantaloupe	Caravelle, Chaparral, Cimaron, Copo deOrio, Cruiser, Early Delight, Gold Rush, Impak, Hy-Mark, Mission, Primo, Ovation, Progresso, Super 45 Open Pollinated: TAMUvalde, Perlita, Mainstream
Carrot	Big Shot,,Candy Stix, Caropak, Cheyenne, Choctaw, Navajo, Sugar Snax, Vita-Sweet, CR 7180, SCR 7248
Cauliflower	Candid Charm Gaudian, Imperial 10-6, Incline, Minuteman, Snowball Y Imp., Snow Crown, White magic, Snowman Green type: Alverde, Macerata, Green Harmony, Spiral Point
Celery	Florida 683, Rocket, Starlet, Summer or Giant Pascal, Utah 52-70
Collard	Champion, Flash, Top Bunch, Vates
Cowpea (Southern Pea)	Pinkeye: Texas Pinkeye, Purple Hull, Pinkeye Purple Hull BVR, Pinkeye Purple Hull, Coronett Blackeye: Blackeye #5, Arkansas #1, Blackeye #46 Cream: Cream 40 Crowder: Brown Sugar, Mississippi Silver, Zipper Greenhouse: Bruneva, Brunex, Vitomil
Cucumber	Slicer: Conquistador, Dasher II, Daytona, General Lee, Indy, Panther, Pointsett 76, Raider, Slice Master, Slice Nice, Supersett, Sprint 440 II, Tunderbird, Turbo Pickling: Calypso, Carolina, Fancypak M, Flurry, Jackson, Royal
Eggplant	Black Bell, Black Magic, Epic, Classic, Florida High Bush, Florida Market, Night Shadow Oriental type: Ichibon, Tycoon
Garlic	Soft neck type: California Early, California Late, Mexican Pink, Creole, Texas White Eggplant type: (not true garlics): Oriental garlic Hardneck type: Roja, German Red, Valencia
Lettuce	Crisp Head: Great Lakes 659 MT, Mission Loose leaf: Flame, Grand Rapids, Prizehead, Red Sails, Salad Bowl, Two Star, Waldeman’s Green

	Butter head: Buttercrunch Romaine: Valmine, Paris Island
Honeydew Melon	Honeybrew, Megabrew, Morning Ice
Kale	Blue Armor, Blue Arrow, Blue Knight, Dwarf Scotch, Vates, Imp. Dwarf Siverian
Mustard Green	Green Wave, Tendergreen, Southern Giant Curl, Florida Broadleaf
Okra	Clemson Spineless, Lee, Emerald, Clemson 80, Green Best, Cajun Delight, Lee Compact Type: Annie Oakley, Prelude, Blondy
Onion	Short Day: *Yellow*– Chula Vista, Cougar, Diamante, Encino, Jaguar, Linda Vista, Marquesa, Mercedes, Riojas, Sweet Sunrise, TX 1015, 6996 *Red*– Rio, Rio Santiago *White*– Diamante, Krystal, Texas Early White Intermediate Day: *Yellow*- Caballero, Cimarron , Riviera , Sierra Blanca, Utopia, Yula *Red*– Fuego *White*– Alabaster , Duro, Spano Long Day: *Yellow*– Armada, Blanco Duro, Capri, Durango, El Charo, Ole', Seville, Sweet Perfection, Valdez:, Vega, Vaquero *Red*– Tango *White*– Sterling
Pepper	Bell: Aladdin, Capistrano, Camelot X3R, Early Sunsation, Jupiter, Pip, Red Knight, Summersweet 840, Taurus, Valiant, X3R Wizard Jalapeno: Coyama, Grande, Mitla, Ole~, Perfecto, TAM Mild-1, TAM Veracruz, Tula, Tulleon Summer Heat 5000, Summer Heat 6000, X3R-Ixtapa, Spp 7603 Serrano: Fiesta, Tampico, Tuxtlas
Potato	Russet: Russet Norkatah, Norgold M, Century Russet White: Alantic, Gemchip, Chipeta, Kennebec Red: Red LaSoda, Viking, Pontiac Yellow Flesh: Yukon Gold
Pumpkin	Mini: Jack-Be-Little, Muchkin, Pro Gold 100 Small: Small Sugar, Triple Treat, Streaker, Pro Gold 300, Oz, Spookie Large: Appalachin, Connecticut Field, Ghost Rider, Howden, Happy Jack, Magic, Pro Gold 500, Pro Gold 510, Trickster, Wizard Mammoth: Alantic, Giant, Big Mac, Big Max, Howden Biggie, Prizewinner
Spinach	Fresh: Fall Green, Samish,, Winter Green (Ark 88-310) Processing: ACX 5044, F 380, ACX 3633, ACX 2615, 6710157
Squash (Yellow)	Straight Neck: General Patton, Golden Girl, Goldbar, Gold Spike, Lemon Drop L., Multipik, PS- 391 Crook Neck: Bandit, Dixie, Early Golden, Freedom II, Goldslice, Goldie, Liverator III, Medallion, Meigs, Prelude II, Pavo, Supersett, Sunrise Zucchini: Commander, Enterprise, Independence II, President, Senator, ACX 34
Sweet Corn	Standard: Merit Y, Jubilee Y, Silver Queen W Se: Calico Belle B, Guadalupe Gold, Kandy Korn, Snowbelle w, Sweet G-90 B, Temptation B Sh2: Challenger Y, Dazzle, Even Sweeter W, Endeavor Y, Florida

	Staysweet Y, Pounchline Y, Summersweet 7710 Y, Sweetie 82 Y, Frontier W, Summersweet 7211 W, Summersweet 7210Y, Summersweet 8102 B Se X Sh2: Sweet Ice, Sweet Symphony, Sweet Rhythm
Sweet Potato	Orange flesh: Beuregard, Jewel, Excel, Hernandez Gold flesh: Shore Gold White flesh: Sumor, White Delight
Tomato	Standard: Bingo, Carnival, Celebrity, Florida 51, Merced, Sanibel, Spitfire, Sunbeam, Sunrise, Summer Flavor 5000 Heat set: ACX 12, Florida 91, Florasette, Heatwave, Sunchaser, Surefire Processing: ACX 8625, Aztec,Casa Del Sol, Chico III, Ohio 8245, TX III, XP 671, Yaqui
Turnip	Greens: All Top, Alamo, Topper Roots: Purple Top White Globe, Royal Globe, Shogoin, York, Seven Top, Tokyo Cross, White Lady, Royal Crown
Watermelon	Hybrids: Big Stripe, Royal Sweet, Royal Flush, Sentinel, Stargazer, Stars-N-Stripes, Summer Flavor 800, Summergold Y Seedless(Triploids): Crimson Trio, Tri X313, Caurosel, Revolution, Summer Flavor 5244 Open Pollinated: Allsweet, JubileeII, Legacy

Table 3. Fertilizer Requirements of Selected Vegetable Crops.

	Generalized Requirement Range			**Apply when Soil Test is very low**		
Crop	**N**	**P**	**K**	**N**	**P**	**K**
Green Bean	60-80	80-100	60-80	75	90	100
Pinto Bean	40-60	80-120	80-120	70	80	100
Beets	60-80	60-80	80-100	100	80	100
Cantaloupe	40-100	60-100	60-100	120	80	120
Carrot	30-80	60-100	60-100	100	80	100
Cauliflower	60-100	120-170	60-80	150	90	120
Swiss Chard	120	80-100	100-120	120	90	160
Collards & Kale	60-100	80-100	80-100	150	90	150
Cucumber	40-100	90-100	140-150	120	80	120
Eggplant	70-115	90-120	150-200	145	90	160
Lettuce	60-90	100-120	120-170	100	80	120
Mustard	60-90	80-120	120-170	100	80	120
Okra	60-80	60-80	120-170	80	70	90
Onion (Dry bulb)	60-90	60-70	100-170	120	80	100
Onion (Green)	60-80	80-120	60-80	100	80	100
Pea (Southern)	20-60	50-100	40-120	60	50	40
Pea (English)	20-80	50-80	40-150	20	50	40
Pepper	30-80	50-80	80-120	160	80	150

Potato (Irish)	80-160	80-120	80-120	180	100	200
Potato (Sweet)	40-60	50-120	120-180	80	90	160
Pumpkin	60-80	80-100	110-140	120	80	140
Radish	40-60	50-80	70-120	90	50	100
Spinach	60-70	110-140	110-140	150	80	125
Squash (Summer)	60-70	65-140	110-140	100	65	120
Squash (Winter)	60-70	100-140	110-140	110	80	125
Sweet Corn	120	70-120	80-100	140	85	100
Tomato	60-80	60-70	110-140	150	80	150
Turnip (Roots)	36-80	50-120	70-120	80	50	100
Turnip (Tops)	36-80	80-120	70-120	80	50	100
Watermelon	40-90	40-60	40-60	120	75	130

TAEX recommendations

Table 4. Approximate Absorption of Nutrients by Traditional Vegetable Crops.

Vegetable	**Yield(cwt./acre)**	**Nutrient Absorption (lb./acre)**		
		N	**P**	**K**
Bean, green	100 beans Plants	120 50 170	10 6 16	55 45 100
Broccoli	100 heads Other	20 145 165	2 8 10	45 165 210
Brussels sprouts	160 sprouts Other	150 85 235	20 9 29	125 110 235
Carrot	500 roots Tops	80 65 145	20 5 25	200 145 345
Celery	1000 tops Roots	170 25 195	35 15 50	380 55 435
Corn, sweet	130 ears Plants	55 100 155	8 12 20	30 75 105
Honeydew melon	290 fruits Vines	70 135 205	8 15 23	65 95 160
Lettuce	350 plants	95	12	170
Muskmelon	225 fruits Vines	95 60 155	17 8 25	120 35 155

Onion	400 bulbs Tops	110 35 145	20 5 25	110 45 155
Pepper	225 fruits Plants	45 95 140	6 6 12	50 90 140
Pea, shelled	40 peas Vines	100 70 170	10 12 22	30 50 80
Potato	400 tubers Vines	150 60 210	19 11 30	200 75 275
Spinach	200 Plants	100	12	100
Sweet Potato	300 roots Vines	80 60 140	16 4 20	160 40 200
Tomato	600 fruits	100 80 180	10 11 21	180 100 280

Source: Knott’s Vegetable Growers Handbook. 4th edition. Don Maynard and George Hochmut.

Table 5. Typical Composition of Manures and Other Organic Fertilizer Sources.

Manures vary greatly in their nutrient content according to the kind of feed used, the percentage and type of litter or bedding, the moisture content, and the age and degree of rotting or drying.. The following data are representative analyses from several reports.

	Moisture %	**Approximate Composition (lb/ton)**		
		N	P2O5	K2O
Fresh Manure with Bedding or Litter				
Cow	86	11	4	10
Duck	61	22	29	10
Goose	67	22	11	10
Hen	73	22	22	10
Hog	87	11	6	9
Horse	80	13	5	13
Sheep	70	20	15	21
Steer or feed yard	85	12	7	11
Turkey	74	26	14	10
Dried Commercial Products				
Cow	21	20	20	38
Hen	13	31	35	40
Hog	10	45	42	20
Rabbit	16	26	31	32
Sheep	10	32	25	41

Stockyard	17	25	24	42
Alfalfa hay	10	50	11	50
Alfalfa straw	7	28	7	36
Barley hay	9	23	11	33
Barley straw	10	12	5	32
Bean straw	11	20	6	25
Beggarweed hay	9	50	12	56
Buckwheat straw	11	14	2	48
Clover hay Alyce Bur Crimson Ladino Sweet	11 8 11 12 8	35 60 45 60 60	21 11 13 12	70 67 67 38
Cowpea hay	10	60	13	36
Cowpea straw	9	20	5	38
Field pea hay	11	28	11	30
Horse bean hay	9	43		
Lezpedeza hay	11	41	8	22
Lezpedeza straw	10	21		
Oat hay	12	26	9	20
Oat straw	10	13	5	33
Ryegrass hay	11	26	11	25
Rye hay	9	21	8	25
Rye straw	7	11	4	22
Sorghum stover, Hegari	13	18	4	
Soybean hay	12	46	11	20
Soybean straw	11	13	6	15
Sudan grass hay	11	28	12	31
Sweet corn fodder	12	30	8	24
Velvet bean hay	7	50	11	53
Vetch hay Common Hairy	11 12	43 62	15 15	53 47
Wheat hay	10	20	8	35
Wheat straw	8	12	3	19
Miscellaneous Organic Materials				
Bat guano		200	80	40
Blood		260	40	20

Bone meal, raw		60	440	
Bone meal, steamed		20	300	
Castor bean meal		100	40	20
Cotton seed meal		120	60	30
Fish meal		200	120	
Garbage tankage		50	40	20
Peanut meal		140	30	24
Sewage sludge		30	25	8
Sewage sludge, act.		120	60	4
Soybean meal		140	24	30
Tankage		140	200	30

Table 6. Crop sensitivity to soil pH.

Crop	Ideal pH	Acceptable Range	Slight alkaline tolerance (pH>7.5)	Slight acid tolerance (6.8-6.0)	Moderate acid tolerance (6.8-5.5)	Strong acid tolerance (6.8-5.0)
Asparagus	6.5	6.0-7.5		X		
Bean	6.5	5.5-6.8		X	X	
Beet	6.5	6.0-8.0	X	X		
Broccoli	6.5	6.0-7.5		X		
Cabbage	6.5	6.0-7.5		X		
Cantaloupe	6.5	6.0-8.0	X	X		
Carrot	6.5	5.5-7.8	X		X	
Cauliflower	6.5	6.0-7.8	X		X	
Celery	6.5	6.0-7.5			X	
Collard	6.5	6.0-7.5			X	
Cowpea (Southern pea)	6.5	6.0-7.5		X		
Cucumber	5.5	6.3-7.5	X		X	
Eggplant	6.5	5.5-7.2			X	
Garlic	6.5	6.0-8.4	X		X	
Lettuce	6.5	6.0-7.6	X	X		
Mustard green	6.3	5.5-7.5			X	
Okra	6.5	6.0-7.5		X		
Onion	6.3	6.0-8.4	X	X		
Pepper	6.5	5.5-7.5			X	
Potato	6	5.0-7.8	X			X
Pumpkin	6.5	5.5-7.5			X	
Radish	6.5	6.0-7.0			X	

Spinach	7	6.5-8.0	X	X		
Squash	6.5	6.0-7.5			X	
Sweet corn	6.5	6.0-7.0			X	
Sweet potato	6.5	5.0-7.5				X
Tomato	6.5	5.5-7.3			X	
Turnip	6.5	5.5-7.5			X	
Watermelon	6	6.5-7.0		X		X

Table 7. Texas Department of Agriculture approved materials list.

Allowed	Allowed with Restrictions
Acetic acid	Alcohol (synthetic)
Adhesives (natural)	Ammonium carbonate
Agar (carageenen kombu, nori)	Ammonium soaps
Alcohol (natural)	Antibiotics
Alfalfa meal	Arsenic (copper chromic arsenate)
Alginated	Ash (natural)
Antitranspirants (natural)	Bleach (sodium hypochlorite and calcium hypochlorite)
Ascorbic acid	Blood meal
Attapulgite clay (Fuller’s Earth)	Bone meal
Bacillus thuringiensis (Bt)	Bordeaux mixtures
Ballons & other inflatable traps	Botanical insecticides
Barriers or repellents	Calcium chloride
Banana oil	Calcium lignosulfate (ligninosulfonate)
Beeswax	Caustic soda
Beneficial organisms	Chelates
Biodynamic preparations	Chilean nitrae (mined nitrate of soda and Chile saltpeter)
Biological controls	Cocoa bean hulls (cocoa shell meal)
Bird traps or netting (mechanical)	Copper hydroxide
Blue-green algae	Corn calcium
Borax (sodium borate)	Cotton gin trash
Boric acid	Cottonseed meal
Boron products	Deer and rabbit repellents
Calcium carbonate (aragonite)	Epsom salts (magnesium sulfate)
Calcium phosphate (dibasic, monobasic, & tribasic)	Fish emulsions
Calcium sulfate (anhydrite and gypsum)	Fish meal
Carbon dioxide gas	Fruit waxes (natural)
Citric acid	Fulvic acid

Citrus oil	Gibberelic acid (sugar carrier based)
Coconut oil	Hydrated lime (slake lime)
Composts	Ligninosulfonate
Compost tea	Lime (calcium hydroxide, caustic lime)
Copper	Lime sulfur (calciumpolysulfide)
Copper sulfate	Magnesium chloride
Corn starch	Micronutrient sprays
Cytokinins	Muriate of potash (potassium chloride)
Detergents	Mushroom compost
Diatomaceous earth	Neem extracts (powder and seeds)
Dolomite	Nematocides (sea animal based)
Dormant oils	Niter
Earthworm castings	Nitrate of soda-potash
Enzymes (natural, amylase, protease, lipase and cellulase)	Nitrogen gas
Feather meal	Oleic acid
Feldspar (potassium aluminosilicate)	Ortho-phosphoric acid
Fungi (entomopathic)	Oxalic acid
Garlic	Petroleum distillates (mineral oils & parafinic oil)
Gelatin waxes	Petroleum oil spray adjuvants (spreader-stickers and carriers)
Glycerin	Pheromones (mating disruption)
Granite dust	Piperonyl butoxide (PBO)
Grape & other pomaces	Polyvinyl alcohol
Greensand (glauconite)	Potassium sulfate
Growth enhancers	Pyrethrums
Guano (bat or bird)	Quassia
Gums (natural)	Rotenone
Gums	Ryania
Gypsum	Sabadilla
Herbal preparations	Seeds (treated with synthetic materials)
Hoof and horn meal	Soda ash
Humates	Sodium molybdate
Humic acid derivatives	Sodium nitrate
Hydrogen peroxide	Sodium silicate

Insect extracts	Stearic acid
Insect feeding stimulants	Sugar beet lime
Kelp extracts	Sulfate of potash
Kelp meal	Sulfates of zinc or iron
Kiesertie (magnesium sulfate and epsomite)	Sulfur (brimstone or flowers of sulfur)
Kiln dust	Transplants (non-certified organically grown)
Langbeinite	Vitamin D3
Lignite	Wood ash
Lime & fluid limestone	Zeolite (natural)
Magnesium carbonate (magnesite)	
Malic acid	
Manures (animal)	
Manure tea	
Marl	
Mechanical and cultural controls	
Microbial diseases	
Microbial plant, soil, compost and seed inoculants	
Mined minerals	
Mineral salts	
Mulches	
Nematocides (natural)	
Nematodes	
Newspaper mulch	
Nosema species	
Nuclear polyhedrosis virus (NPV)	
Oystershell lime	
Peanut meal	
Peat moss	
Pectin	
Perlite	
pH buffers (natural)	
Pheromones (monitoring)	

Phosphate rock (phosphorite)	
Pine (tall) oil	
Plant extracts	
Potato starch	
Propolis	
Protozoa	
Rodent traps	
Rice hulls	
Salt	
Saw dust	
Seaweed and seaweed extracts	

Note: Periodically check with TDA to make certain products are still on approved list.

Table 8. The National List of Allowed and Prohibited Substances

§ 205.600 Evaluation criteria for allowed and prohibited substances, methods, and ingredients.
The following criteria will be utilized in the evaluation of substances or ingredients for the organic production and handling sections of the National List:

(a) Synthetic and nonsynthetic substances considered for inclusion on or deletion from the National List of allowed and prohibited substances will be evaluated using the criteria specified in the Act (7 U.S.C. 6517 and 6518).
(b) In addition to the criteria set forth in the Act, any synthetic substance used as a processing aid or adjuvant will be evaluated against the following criteria:

(1) The substance cannot be produced from a natural source and there are no organic substitutes;
(2) The substance's manufacture, use, and disposal do not have adverse effects on the environment and are done in a manner compatible with organic handling;
(3) The nutritional quality of the food is maintained when the substance is used, and the substance, itself, or its breakdown products do not have an adverse effect on human health as defined by applicable Federal regulations;
(4) The substance's primary use is not as a preservative or to recreate or improve flavors, colors, textures, or nutritive value lost during processing, except where the replacement of nutrients is required by law;
(5) The substance is listed as generally recognized as safe (GRAS) by Food and Drug Administration (FDA) when used in accordance with FDA's good manufacturing practices (GMP) and contains no residues of heavy metals or other contaminants in excess of tolerances set by FDA; and
(6) The substance is essential for the handling of organically produced agricultural products.

(c) Nonsynthetics used in organic processing will be evaluated using the criteria specified in the Act (7 U.S.C. 6517 and 6518).

§ 205.601 Synthetic substances allowed for use in organic crop production.
In accordance with restrictions specified in this section, the following synthetic substances may be used in organic crop production:

(a) As algicide, disinfectants, and sanitizer, including irrigation system cleaning systems

(1) Alcohols

(i) Ethanol
(ii) Isopropanol

(2) Chlorine materials – Except, That, residual chlorine levels in the water shall not exceed the maximum residual disinfectant limit under the Safe Drinking Water Act.

(i) Calcium hypochlorite
(ii) Chlorine dioxide
(iii) Sodium hypochlorite

(3) Hydrogen peroxide
(4) Soap-based algicide/demisters

(b) As herbicides, weed barriers, as applicable.

(1) Herbicides, soap-based – for use in farmstead maintenance (roadways, ditches, right of ways, building perimeters) and ornamental crops
(2) Mulches

(i) Newspaper or other recycled paper, without glossy or colored inks.
(ii) Plastic mulch and covers (petroleum-based other than polyvinyl chloride (PVC))

(c) As compost feedstocks

Newspapers or other recycled paper, without glossy or colored inks

(d) As animal repellents

Soaps, ammonium – for use as a large animal repellant only, no contact with soil or edible portion of crop

(e) As insecticides (including acaricides or mite control)

(1) Ammonium carbonate – for use as bait in insect traps only, no direct contact with crop or soil
(2) Boric acid – structural pest control, no direct contact with organic food or crops
(3) Elemental sulfur
(4) Lime sulfur – including calcium polysulfide
(5) Oils, horticultural – narrow range oils as dormant, suffocating, and summer oils.
(6) Soaps, insecticidal
(7) Sticky traps/barriers

(f) As insect attractants

Pheromones

(g) As rodenticides

(1) Sulfur dioxide – underground rodent control only (smoke bombs)
(2) Vitamin D3

(h) As slug or snail bait

<None>

(i) As plant disease control

(1) Coppers, fixed – copper hydroxide, copper oxide, copper oxychloride, includes products exempted from EPA tolerance, Provided, That, copper-based materials must be used in a manner that minimizes accumulation in the soil and shall not be used as herbicides.
(2) Copper sulfate – Substance must be used in a manner that minimizes accumulation of copper in the soil.
(3) Hydrated lime – must be used in a manner that minimizes copper accumulation in the soil.
(4) Hydrogen peroxide
(5) Lime sulfur
(6) Oils, horticultural, narrow range oils as dormant, suffocating, and summer oils.
(7) Potassium bicarbonate
(8) Elemental sulfur
(9) Streptomycin, for fire blight control in apples and pears only
(10) Tetracycline (oxytetracycline calcium complex), for fire blight control only

(j) As plant or soil amendments.

(1) Aquatic plant extracts (other than hydrolyzed) – Extraction process is limited to the use of potassium hydroxide or sodium hydroxide; solvent amount used is limited to that amount necessary for extraction.
(2) Elemental sulfur
(3) Humic acids – naturally occurring deposits, water and alkali extracts only
(4) Lignin sulfonate – chelating agent, dust suppressant, floatation agent
(5) Magnesium sulfate – allowed with a documented soil deficiency
(6) Micronutrients – not to be used as a defoliant, herbicide, or desiccant. Those made from nitrates or chlorides are not allowed. Soil deficiency must be documented by testing.

(i) Soluble boron products
(ii) Sulfates, carbonates, oxides, or silicates of zinc, copper, iron, manganese, molybdenum, selenium, and cobalt,

(7) Liquid fish products – can be pH adjusted with sulfuric, citric or phosphoric acid. The amount of acid used shall not exceed the minimum needed to lower the pH to 3.5
(8) Vitamins, B1, C, and E

(k) As plant growth regulators

Ethylene – for regulation of pineapple flowering

(l) As floating agents in postharvest handling

(1) Lignin sulfonate
(2) Sodium silicate – for tree fruit and fiber processing

(m) As synthetic inert ingredients as classified by the Environmental Protection Agency (EPA), for use with nonsynthetic substances or synthetic substances listed in this section and used as an active pesticide ingredient in accordance with any limitations on the use of such substances.

(1) EPA List 4 – Inerts of Minimal Concern

(n)-(z) [Reserved]

§ 205.602 Nonsynthetic substances prohibited for use in organic crop production.
The following nonsynthetic substances may not be used in organic crop production:

(a) Ash from manure burning
(b) Arsenic
(c) Lead salts
(d) Sodium fluoaluminate (mined)
(e) Strychnine
(f) Tobacco dust (nicotine sulfate)
(g) Potassium chloride – unless derived from a mined source and applied in a manner that minimizes chloride accumulation in the soil.
(h) Sodium nitrate – unless use is restricted to no more than 20% of the crop's total nitrogen requirement.
(i)-(z) [Reserved]

§ 205.603 Synthetic substances allowed for use in organic livestock production.
In accordance with restrictions specified in this section the following synthetic substances may be used in organic livestock production:

(a) As disinfectants, sanitizer, and medical treatments as applicable

(1) Alcohols

(i) Ethanol – disinfectant and sanitizer only, prohibited as a feed additive
(ii) Isopropanol – disinfectant only

(2) Aspirin – approved for health care use to reduce inflammation
(3) Chlorine materials – disinfecting and sanitizing facilities and equipment. Residual chlorine levels in the water shall not exceed the maximum residual disinfectant limit under the Safe Drinking Water Act

(i) Calcium hypochlorite
(ii) Chlorine dioxide
(iii) Sodium hypochlorite

(4) Chlorohexidine – Allowed for surgical procedures conducted by a veterinarian. Allowed for use as a teat dip when alternative germicidal agents

and/or physical barriers have lost their effectiveness
(5) Electrolytes – without antibiotics
(6) Glucose
(7) Glycerin – Allowed as a livestock teat dip, must be produced through the hydrolysis of fats or oils
(8) Iodine
(9) Hydrogen peroxide
(10) Magnesium sulfate
(11) Oxytocin – use in postparturition therapeutic applications
(12) Parasiticides

Ivermectin – prohibited in slaughter stock, allowed in emergency treatment for dairy and breeder stock when organic system plan-approved preventive management does not prevent infestation. Milk or milk products from a treated animal cannot be labeled as provided for in subpart D of this part for 90 days following treatment. In breeder stock, treatment cannot occur during the last third of gestation if the progeny will be sold as organic and must not be used during the lactation period of breeding stock.

(13) Phosphoric acid – allowed as an equipment cleaner, Provided, That, no direct contact with organically managed livestock or land occurs.
(14) Biologics

Vaccines

(b) As topical treatment, external parasiticide or local anesthetic as applicable.

(1) Iodine
(2) Lidocaine – as a local anesthetic. Use requires a withdrawal period of 90 days after administering to livestock intended for slaughter and 7 days after administering to dairy animals
(3) Lime, hydrated – (bordeaux mixes), not permitted to cauterize physical alterations or deodorize animal wastes.
(4) Mineral oil – for topical use and as a lubricant
(5) Procaine – as a local anesthetic, use requires a withdrawal period of 90 days after administering to livestock intended for slaughter and 7 days after administering to dairy animals
(6) Copper sulfate

(c) As feed supplements

Milk replacers – without antibiotics, as emergency use only, no nonmilk products or products from BST treated animals

(d) As feed additives

(1) Trace minerals, used for enrichment or fortification when FDA approved, including:

(i) Copper sulfate
(ii) Magnesium sulfate

(2) Vitamins, used for enrichment or fortification when FDA approved

(e) As synthetic inert ingredients as classified by the Environmental Protection Agency (EPA), for use with nonsynthetic substances or a synthetic substances listed in this section and used as an active pesticide ingredient in accordance with any limitations on the use of such substances.
EPA List 4 – Inerts of Minimal Concern.
(f)-(z) [Reserved]

§ 205.604 Nonsynthetic substances prohibited for use in organic livestock production.
The following nonsynthetic substances may not be used in organic livestock production:

(a) Strychnine
(b)-(z) [Reserved]

§ 205.605 Nonagricultural (nonorganic) substances allowed as ingredients in or on processed products labeled as "organic" or "made with organic (specified ingredients or food group(s))."
The following nonagricultural substances may be used as ingredients in or on processed products labeled as "organic" or "made with organic (specified ingredients or food group(s))" only in accordance with any restrictions specified in this section.

(a) Nonsynthetics allowed:

(1) Acids

(i) Alginic
(ii) Citric – produced by microbial fermentation of carbohydrate substances
(iii) Lactic

(2) Bentonite
(3) Calcium carbonate
(4) Calcium chloride
(5) Colors, nonsynthetic sources only
(6) Dairy cultures
(7) Diatomaceous earth – food filtering aid only
(8) Enzymes – must be derived from edible, nontoxic plants, nonpathogenic fungi, or nonpathogenic bacteria
(9) Flavors, nonsynthetic sources only and must not be produced using synthetic solvents and carrier systems or any artificial preservative.
(10) Kaolin
(11) Magnesium sulfate, nonsynthetic sources only
(12) Nitrogen – oil-free grades
(13) Oxygen – oil-free grades
(14) Perlite – for use only as a filter aid in food processing
(15) Potassium chloride
(16) Potassium iodide
(17) Sodium bicarbonate
(18) Sodium carbonate
(19) Waxes – nonsynthetic

(i) Carnauba wax
(ii) Wood resin

(20) Yeast – nonsynthetic, growth on petrochemical substrate and sulfite waste liquor is prohibited

(i) Autolysate
(ii) Bakers
(iii) Brewers
(iv) Nutritional
(v) Smoked – nonsynthetic smoke flavoring process must be documented.

(b) Synthetics allowed:

(1) Alginates
(2) Ammonium bicarbonate – for use only as a leavening agent
(3) Ammonium carbonate – for use only as a leavening agent
(4) Ascorbic acid
(5) Calcium citrate
(6) Calcium hydroxide
(7) Calcium phosphates (monobasic, dibasic, and tribasic)
(8) Carbon dioxide
(9) Chlorine materials – disinfecting and sanitizing food contact surfaces, Except, That, residual chlorine levels in the water shall not exceed the maximum residual disinfectant limit under the Safe Drinking Water Act.

(i) Calcium hypochlorite
(ii) Chlorine dioxide
(iii) Sodium hypochlorite

(10) Ethylene – allowed for postharvest ripening of tropical fruit
(11) Ferrous sulfate – for iron enrichment or fortification of foods when required by regulation or recommended (independent organization)
(12) Glycerides (mono and di) – for use only in drum drying of food
(13) Glycerin – produced by hydrolysis of fats and oils
(14) Hydrogen peroxide
(15) Lecithin – bleached
(16) Magnesium carbonate – for use only in agricultural products labeled "made with organic (specified ingredients or food group(s))," prohibited in agricultural products labeled "organic"
(17) Magnesium chloride – derived from sea water
(18) Magnesium stearate – for use only in agricultural products labeled "made with organic (specified ingredients or food group(s))," prohibited in agricultural products labeled "organic"
(19) Nutrient vitamins and minerals, in accordance with 21 CFR 104.20, Nutritional Quality Guidelines For Foods
(20) Ozone
(21) Pectin (low-methoxy)
(22) Phosphoric acid – cleaning of food-contact surfaces and equipment only
(23) Potassium acid tartrate
(24) Potassium tartrate made from tartaric acid
(25) Potassium carbonate
(26) Potassium citrate
(27) Potassium hydroxide – prohibited for use in lye peeling of fruits and vegetables
(28) Potassium iodide – for use only in agricultural products labeled "made with

organic (specified ingredients or food group(s)),” prohibited in agricultural products labeled “organic”
(29) Potassium phosphate – for use only in agricultural products labeled “made with organic (specific ingredients or food group(s)),” prohibited in agricultural products labeled “organic”
(30) Silicon dioxide
(31) Sodium citrate
(32) Sodium hydroxide – prohibited for use in lye peeling of fruits and vegetables
(33) Sodium phosphates – for use only in dairy foods
(34) Sulfur dioxide – for use only in wine labeled “made with organic grapes,” Provided, That, total sulfite concentration does not exceed 100 ppm.
(35) Tocopherols – derived from vegetable oil when rosemary extracts are not a suitable alternative
(36) Xanthan gum

(c)-(z) [Reserved]

§ 205.606 Nonorganically produced agricultural products allowed as ingredients in or on processed products labeled as organic or made with organic ingredients.

The following nonorganically produced agricultural products may be used as ingredients in or on processed products labeled as “organic” or “made with organic (specified ingredients or food group(s))” only in accordance with any restrictions specified in this section.

Any nonorganically produced agricultural product may be used in accordance with the restrictions specified in this section and when the product is not commercially available in organic form.

(a) Cornstarch (native)
(b) Gums – water extracted only (arabic, guar, locust bean, carob bean)
(c) Kelp – for use only as a thickener and dietary supplement
(d) Lecithin – unbleached
(e) Pectin (high-methoxy)

§ 205.607 Amending the National List.

(a) Any person may petition the National Organic Standard Board for the purpose of having a substance evaluated by the Board for recommendation to the Secretary for inclusion on or deletion from the National List in accordance with the Act.
(b) A person petitioning for amendment of the National List should request a copy of the petition procedures from the USDA at the address in § 205.607(c).
(c) A petition to amend the National List must be submitted to: Program Manager, USDA/AMS/TMP/NOP, Room 2945, South Building, P.O. Box 96456, Washington, DC 20090-6456.